Software Requirements Specification

For

<u>Hotel Management System</u>

Version 1.0

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1. Introduction

1.1 Purpose

This document describes the requirements for the Hotel Management System (HMS), Version 1.0.
The system is used to manage hotel operations like reservations, check-in/check-out, and billing.
This SRS covers the full system.

1.2 Document Conventions

Requirements are numbered (e.g., REQ-1, REQ-2).
Headings follow a structured format (1, 1.1, etc.).
Important terms may be in bold.
Requirements are written clearly and simply.

1.3 Intended Audience and Reading Suggestions

This document is for developers, testers, project managers, and hotel staff.
Readers should start with the Introduction, then move to the Overall Description, and then System Features.

1.4 Product Scope

The Hotel Management System helps manage hotel tasks such as room booking, customer data, check-in/check-out, and billing.

The main goals are to save time, reduce errors, and improve service.

1.5 References

IEEE SRS Standard
UML Documentation
Hotel system examples

2. Overall Description

2.1 Product Perspective

The Hotel Management System is a standalone system designed to manage hotel operations digitally. It can replace manual or outdated systems currently used in hotels.
The system may also connect with external systems like payment gateways or online booking platforms.

2.2 Product Functions

The system allows users to perform key operations such as managing room bookings, handling check-in and check-out, storing customer information, generating bills, and viewing reports.
It also supports basic administrative tasks like managing rooms and users.

2.3 User Classes and Characteristics

The main users of the system are hotel staff and administrators.
Receptionists use the system frequently for bookings and check-in/check-out.
Administrators manage system settings, users, and reports.
Managers use the system to view reports and monitor performance.
Different users have different access levels based on their roles.

2.4 Operating Environment

The system will run on standard computers or laptops.
It can work on common operating systems such as Windows, Linux, or macOS.
It may be web-based and accessed through a browser like Chrome or Edge.

2.5 Design and Implementation Constraints

The system should follow standard programming practices.
It may be limited by hardware performance or internet availability.
Security and data protection must be considered.
The system should be easy to maintain and scalable.

2.6 User Documentation

User documentation will include a user manual and basic help guides.
 the system to داخلIt may also include online help .أثناء الاستخدامguide users

2.7 Assumptions and Dependencies

It is assumed that users have basic computer knowledge.
The system depends on stable hardware and internet connection if it is web-based.
It may depend on external services like payment systems or databases.

3. External Interface Requirements

3.1 User Interfaces

The system provides a simple and user-friendly interface for hotel staff.
Screens include login, dashboard, room booking, check-in/check-out, and billing pages.
Each screen follows a consistent layout with standard buttons like save, edit, delete, and help.
Error messages are clear and easy to understand.

3.2 Hardware Interfaces

The system works with standard hardware such as computers, keyboards, and printers.
It may connect to printers for invoice printing.
No special hardware is required.

3.3 Software Interfaces

The system interacts with a database to store and retrieve data such as customer details and bookings.
It may also connect to external systems like payment gateways.
The system runs on common operating systems and uses standard software tools.

3.4 Communications Interfaces

The system may use internet communication for online bookings or payments.
It works through web browsers using standard protocols like HTTP or HTTPS.
Data should be secure during transfer.

4. System Features

4.1 Room Booking

4.1.1 Description and Priority

This feature allows users to book rooms for customers.
It includes selecting room type, dates, and customer details.
Priority: High

4.1.2 Stimulus/Response Sequences

The user selects the booking option and enters customer and room details.
The system checks room availability.
If available, the system confirms the booking and stores the data.
If not available, the system shows an error message.

4.1.3 Functional Requirements

REQ-1: The system shall allow users to create a new booking.
REQ-2: The system shall check room availability before confirming booking.
REQ-3: The system shall store booking details in the database.
REQ-4: The system shall display an error message if no rooms are available.

4.2 Check-in and Check-out

4.2.1 Description and Priority

This feature allows hotel staff to manage customer check-in and check-out.
Priority: High

4.2.2 Stimulus/Response Sequences

The user selects a booking and chooses check-in.
The system updates room status to occupied.
At check-out, the system calculates the bill and updates room status to available.

4.2.3 Functional Requirements

REQ-5: The system shall allow check-in for booked customers.
REQ-6: The system shall update room status automatically.
REQ-7: The system shall allow check-out and generate the final bill.
REQ-8: The system shall mark the room as available after check-out.

4.3 Billing and Payments

4.3.1 Description and Priority

This feature handles billing and payment processing.
Priority: High

4.3.2 Stimulus/Response Sequences

The user requests the bill for a customer.
The system calculates total cost based on stay duration and services.
The system displays the bill and processes payment.

4.3.3 Functional Requirements

REQ-9: The system shall calculate the total bill automatically.
REQ-10: The system shall generate an invoice.
REQ-11: The system shall support payment recording.
REQ-12: The system shall display payment confirmation.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system should respond to user actions within a few seconds.
Booking, check-in, and billing operations should be processed quickly without delay.

The system should support multiple users at the same time without performance issues.

5.2 Safety Requirements

The system should prevent data loss by saving data automatically.
Backup mechanisms should be available to restore data if needed.
The system should avoid incorrect operations like double booking.

5.3 Security Requirements

The system should require user login with a username and password.
Different users should have different access levels.
Sensitive data such as customer information and payments should be protected.
Data should be securely stored and transferred.

5.4 Software Quality Attributes

The system should be easy to use and understand.
It should be reliable and available when needed.
The system should be easy to maintain and update.
It should work on different devices and environments.

5.5 Business Rules

Only authorized staff can access the system.
Receptionists handle bookings and check-in/check-out.
Administrators manage users and system settings.
Payments must be completed before check-out is finalized.

6. Other Requirements

The system should use a database to store all hotel data such as customers, rooms, and bookings.
It should support basic language settings if needed.
The system must follow basic legal rules for handling customer data and privacy.

Appendix A: Glossary

HMS: Hotel Management System
Booking: Reserving a room for a customer
Check-in: Customer arrival and room assignment
Check-out: Customer leaving and bill payment
Admin: System administrator

Appendix B: Analysis Models

This section may include diagrams such as system flowcharts, data flow diagrams, or database design diagrams to explain the system structure.

Appendix C: To Be Determined List

TBD-1: Final database design
TBD-2: Payment method integration
TBD-3: Detailed user interface design